

fashion_pt_2

September 24, 2025

```
[1]: ## IMPORTANT: On Colab, we expect your homework to be in the cs189 folder  
## Please contact staff if you encounter any problems with installing  
↳dependencies  
import sys  
IS_COLAB = 'google.colab' in sys.modules  
if IS_COLAB:  
    from google.colab import drive  
    drive.mount('/content/drive')  
    %cd /content/drive/MyDrive/cs189/hw/hw1  
    %pip install -r ./requirements.txt  
    !pip install -U kaleido plotly  
    import kaleido  
    kaleido.get_chrome_sync()  
  
import plotly.io as pio  
pio.renderers.default = pio.renderers.default + "+png"
```

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Mounted at /content/drive  
/content/drive/MyDrive/cs189/hw/hw1  
Collecting otter_grader (from -r ./requirements.txt (line 1))  
  Cloning https://github.com/BerkeleyML/otter-grader-189.git (to revision  
master) to /tmp/pip-install-xd0tmq4i/otter-  
grader_6cccf7834547bcb680a68926b510de  
  Running command git clone --filter=blob:none --quiet  
https://github.com/BerkeleyML/otter-grader-189.git /tmp/pip-install-  
xd0tmq4i/otter-grader_6cccf7834547bcb680a68926b510de  
  Resolved https://github.com/BerkeleyML/otter-grader-189.git to commit  
35e43fdf64a55f4b9222d7b531890fb7d5cb498c  
  Installing build dependencies ... done  
  Getting requirements to build wheel ... done  
  Preparing metadata (pyproject.toml) ... done  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages  
(from -r ./requirements.txt (line 2)) (4.67.1)  
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-  
packages (from -r ./requirements.txt (line 3)) (3.10.0)  
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages  
(from -r ./requirements.txt (line 4)) (5.24.1)  
Requirement already satisfied: numpy>=1.21.0 in /usr/local/lib/python3.12/dist-
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packages (from -r ./requirements.txt (line 5)) (2.0.2)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 6)) (2.2.2)
Requirement already satisfied: scikit-learn>=0.24.0 in
/usr/local/lib/python3.12/dist-packages (from -r ./requirements.txt (line 7))
(1.6.1)
Requirement already satisfied: pytest>=6.0.0 in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 8)) (8.4.2)
Collecting jupyter>=1.0.0 (from -r ./requirements.txt (line 9))
  Downloading jupyter-1.1.1-py2.py3-none-any.whl.metadata (2.0 kB)
Requirement already satisfied: tabulate in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 10)) (0.9.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages
(from -r ./requirements.txt (line 11)) (2.8.0+cu126)
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 12)) (0.23.0+cu126)
Requirement already satisfied: tiktoken in /usr/local/lib/python3.12/dist-
packages (from -r ./requirements.txt (line 13)) (0.11.0)
Requirement already satisfied: click<9.0.0,>=8.1.7 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (8.2.1)
Requirement already satisfied: dill>=0.3.0 in /usr/local/lib/python3.12/dist-
packages (from otter_grader->-r ./requirements.txt (line 1)) (0.3.8)
Collecting fica>=0.4.1 (from otter_grader->-r ./requirements.txt (line 1))
  Downloading fica-0.4.1-py3-none-any.whl.metadata (2.0 kB)
Collecting ipylab<2.0.0,>=1.0.0 (from otter_grader->-r ./requirements.txt (line
1))
  Downloading ipylab-1.1.0-py3-none-any.whl.metadata (6.7 kB)
Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-
packages (from otter_grader->-r ./requirements.txt (line 1)) (7.34.0)
Collecting ipywidgets<9.0.0,>=8.1.5 (from otter_grader->-r ./requirements.txt
(line 1))
  Downloading ipywidgets-8.1.7-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: jinja2<4.0,>=3.1 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (3.1.6)
Requirement already satisfied: jupyter<2.0.0,>=1.16.4 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (1.17.3)
Requirement already satisfied: nbconvert>=6.0.0 in
/usr/local/lib/python3.12/dist-packages (from nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (7.16.6)
Requirement already satisfied: nbformat>=5.0.0 in
/usr/local/lib/python3.12/dist-packages (from otter_grader->-r
./requirements.txt (line 1)) (5.10.4)
Collecting python-on-whales<1.0.0,>=0.72.0 (from otter_grader->-r
./requirements.txt (line 1))

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Downloading python_on_whales-0.78.0-py3-none-any.whl.metadata (18 kB)
 Requirement already satisfied: pyyaml<7,>=6 in /usr/local/lib/python3.12/dist-packages (from otter_grader->-r ./requirements.txt (line 1)) (6.0.2)
 Requirement already satisfied: requests<3.0,>=2.31 in /usr/local/lib/python3.12/dist-packages (from otter_grader->-r ./requirements.txt (line 1)) (2.32.4)
 Requirement already satisfied: wrapt<2.0.0,>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from otter_grader->-r ./requirements.txt (line 1)) (1.17.3)
 Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (1.3.3)
 Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (4.60.0)
 Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (1.4.9)
 Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (25.0)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (3.2.4)
 Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->-r ./requirements.txt (line 3)) (2.9.0.post0)
 Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly->-r ./requirements.txt (line 4)) (8.5.0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->-r ./requirements.txt (line 6)) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->-r ./requirements.txt (line 6)) (2025.2)
 Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (1.16.2)
 Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (1.5.2)
 Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.24.0->-r ./requirements.txt (line 7)) (3.6.0)
 Requirement already satisfied: iniconfig>=1 in /usr/local/lib/python3.12/dist-packages (from pytest>=6.0.0->-r ./requirements.txt (line 8)) (2.1.0)
 Requirement already satisfied: pluggy<2,>=1.5 in /usr/local/lib/python3.12/dist-

packages (from pytest>=6.0.0->-r ./requirements.txt (line 8)) (1.6.0)
 Requirement already satisfied: pygments>=2.7.2 in
 /usr/local/lib/python3.12/dist-packages (from pytest>=6.0.0->-r
 ./requirements.txt (line 8)) (2.19.2)
 Requirement already satisfied: notebook in /usr/local/lib/python3.12/dist-
 packages (from jupyter>=1.0.0->-r ./requirements.txt (line 9)) (6.5.7)
 Requirement already satisfied: jupyter-console in
 /usr/local/lib/python3.12/dist-packages (from jupyter>=1.0.0->-r
 ./requirements.txt (line 9)) (6.6.3)
 Requirement already satisfied: ipykernel in /usr/local/lib/python3.12/dist-
 packages (from jupyter>=1.0.0->-r ./requirements.txt (line 9)) (6.17.1)
 Collecting jupyterlab (from jupyter>=1.0.0->-r ./requirements.txt (line 9))
 Downloading jupyterlab-4.4.7-py3-none-any.whl.metadata (16 kB)
 Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-
 packages (from torch->-r ./requirements.txt (line 11)) (3.19.1)
 Requirement already satisfied: typing-extensions>=4.10.0 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (4.15.0)
 Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
 packages (from torch->-r ./requirements.txt (line 11)) (75.2.0)
 Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-
 packages (from torch->-r ./requirements.txt (line 11)) (1.13.3)
 Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-
 packages (from torch->-r ./requirements.txt (line 11)) (3.5)
 Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages
 (from torch->-r ./requirements.txt (line 11)) (2025.3.0)
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (12.6.77)
 Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (12.6.77)
 Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (12.6.80)
 Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (9.10.2.21)
 Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (12.6.4.1)
 Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (11.3.0.4)
 Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in
 /usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
 11)) (10.3.7.77)
 Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in

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/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (11.7.1.2)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (12.5.4.2)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (2.27.3)
Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (12.6.77)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (12.6.85)
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in
/usr/local/lib/python3.12/dist-packages (from torch->-r ./requirements.txt (line
11)) (1.11.1.6)
Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-
packages (from torch->-r ./requirements.txt (line 11)) (3.4.0)
Requirement already satisfied: regex>=2022.1.18 in
/usr/local/lib/python3.12/dist-packages (from tiktoken->-r ./requirements.txt
(line 13)) (2024.11.6)
Requirement already satisfied: docutils in /usr/local/lib/python3.12/dist-
packages (from fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1))
(0.21.2)
Requirement already satisfied: sphinx in /usr/local/lib/python3.12/dist-packages
(from fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (8.2.3)
Collecting comm>=0.1.3 (from ipywidgets<9.0.0,>=8.1.5->otter_grader->-r
./requirements.txt (line 1))
  Downloading comm-0.2.3-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: traitlets>=4.3.1 in
/usr/local/lib/python3.12/dist-packages (from
ipywidgets<9.0.0,>=8.1.5->otter_grader->-r ./requirements.txt (line 1)) (5.7.1)
Collecting widgetsnbextension~=4.0.14 (from
ipywidgets<9.0.0,>=8.1.5->otter_grader->-r ./requirements.txt (line 1))
  Downloading widgetsnbextension-4.0.14-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: jupyterlab_widgets~=3.0.15 in
/usr/local/lib/python3.12/dist-packages (from
ipywidgets<9.0.0,>=8.1.5->otter_grader->-r ./requirements.txt (line 1)) (3.0.15)
Collecting jedi>=0.16 (from ipython->otter_grader->-r ./requirements.txt (line
1))
  Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
packages (from ipython->otter_grader->-r ./requirements.txt (line 1)) (4.4.2)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-

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packages (from ipython->otter_grader->-r ./requirements.txt (line 1)) (0.7.5)
 Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
 /usr/local/lib/python3.12/dist-packages (from ipython->otter_grader->-r
 ./requirements.txt (line 1)) (3.0.52)
 Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
 packages (from ipython->otter_grader->-r ./requirements.txt (line 1)) (0.2.0)
 Requirement already satisfied: matplotlib-inline in
 /usr/local/lib/python3.12/dist-packages (from ipython->otter_grader->-r
 ./requirements.txt (line 1)) (0.1.7)
 Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
 packages (from ipython->otter_grader->-r ./requirements.txt (line 1)) (4.9.0)
 Requirement already satisfied: MarkupSafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from jinja2<4.0,>=3.1->otter_grader->-r
 ./requirements.txt (line 1)) (3.0.2)
 Requirement already satisfied: markdown-it-py>=1.0 in
 /usr/local/lib/python3.12/dist-packages (from
 jupyter<2.0.0,>=1.16.4->otter_grader->-r ./requirements.txt (line 1)) (4.0.0)
 Requirement already satisfied: mdit-py-plugins in
 /usr/local/lib/python3.12/dist-packages (from
 jupyter<2.0.0,>=1.16.4->otter_grader->-r ./requirements.txt (line 1)) (0.5.0)
 Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-
 packages (from nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform !=
 "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
 (line 1)) (4.13.5)
 Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-
 packages (from bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;
 sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
 ./requirements.txt (line 1)) (6.2.0)
 Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-
 packages (from nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform !=
 "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
 (line 1)) (0.7.1)
 Requirement already satisfied: jupyter-core>=4.7 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
 sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (5.8.1)
 Requirement already satisfied: jupyterlab-pygments in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
 sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (0.3.0)
 Requirement already satisfied: mistune<4,>=2.0.3 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
 sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (3.1.4)
 Requirement already satisfied: nbclient>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
 sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (0.10.2)

Requirement already satisfied: pandocfilters>=1.4.1 in
 /usr/local/lib/python3.12/dist-packages (from
 nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
 sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (1.5.1)
 Collecting playwright (from nbconvert[webpdf]>=6.0.0; sys_platform !=
 "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
 (line 1))
 Downloading playwright-1.55.0-py3-none-manylinux1_x86_64.whl.metadata (3.5 kB)
 Requirement already satisfied: fastjsonschema>=2.15 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.0.0->otter_grader->-r
 ./requirements.txt (line 1)) (2.21.2)
 Requirement already satisfied: jsonschema>=2.6 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.0.0->otter_grader->-r
 ./requirements.txt (line 1)) (4.25.1)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
 packages (from python-dateutil>=2.7->matplotlib->-r ./requirements.txt (line 3))
 (1.17.0)
 Requirement already satisfied: pydantic!=2.0.*,<3,>=2 in
 /usr/local/lib/python3.12/dist-packages (from python-on-
 whales<1.0.0,>=0.72.0->otter_grader->-r ./requirements.txt (line 1)) (2.11.9)
 Requirement already satisfied: charset_normalizer<4,>=2 in
 /usr/local/lib/python3.12/dist-packages (from
 requests<3.0,>=2.31->otter_grader->-r ./requirements.txt (line 1)) (3.4.3)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
 packages (from requests<3.0,>=2.31->otter_grader->-r ./requirements.txt (line
 1)) (3.10)
 Requirement already satisfied: urllib3<3,>=1.21.1 in
 /usr/local/lib/python3.12/dist-packages (from
 requests<3.0,>=2.31->otter_grader->-r ./requirements.txt (line 1)) (2.5.0)
 Requirement already satisfied: certifi>=2017.4.17 in
 /usr/local/lib/python3.12/dist-packages (from
 requests<3.0,>=2.31->otter_grader->-r ./requirements.txt (line 1)) (2025.8.3)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in
 /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->-r
 ./requirements.txt (line 11)) (1.3.0)
 Requirement already satisfied: debugpy>=1.0 in /usr/local/lib/python3.12/dist-
 packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))
 (1.8.15)
 Requirement already satisfied: jupyter-client>=6.1.12 in
 /usr/local/lib/python3.12/dist-packages (from ipykernel->jupyter>=1.0.0->-r
 ./requirements.txt (line 9)) (7.4.9)
 Requirement already satisfied: nest-asyncio in /usr/local/lib/python3.12/dist-
 packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))
 (1.6.0)
 Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages
 (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (5.9.5)
 Requirement already satisfied: pyzmq>=17 in /usr/local/lib/python3.12/dist-
 packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))

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(26.2.1)
Requirement already satisfied: tornado>=6.1 in /usr/local/lib/python3.12/dist-
packages (from ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(6.4.2)
Collecting async-lru>=1.0.0 (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9))
  Downloading async_lru-2.0.5-py3-none-any.whl.metadata (4.5 kB)
Requirement already satisfied: httpx<1,>=0.25.0 in
/usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.28.1)
Collecting jupyter-lsp>=2.0.0 (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9))
  Downloading jupyter_lsp-2.3.0-py3-none-any.whl.metadata (1.8 kB)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in
/usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (2.14.0)
Collecting jupyterlab-server<3,>=2.27.1 (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9))
  Downloading jupyterlab_server-2.27.3-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: notebook-shim>=0.2 in
/usr/local/lib/python3.12/dist-packages (from jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.2.4)
Requirement already satisfied: argon2-cffi in /usr/local/lib/python3.12/dist-
packages (from notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(25.1.0)
Requirement already satisfied: ipython-genutils in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.2.0)
Requirement already satisfied: Send2Trash>=1.8.0 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.8.3)
Requirement already satisfied: terminado>=0.8.3 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.18.1)
Requirement already satisfied: prometheus-client in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (0.22.1)
Requirement already satisfied: nbclassic>=0.4.7 in
/usr/local/lib/python3.12/dist-packages (from notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.3.3)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
packages (from
bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from
bleach[css]!=5.0.0->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform !=

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"emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (1.4.0)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (4.10.0)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (1.0.9)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.16.0)

Requirement already satisfied: parso<0.9.0,>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->otter_grader->-r ./requirements.txt (line 1)) (0.8.5)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1)) (25.3.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1)) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1)) (0.36.2)

Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat>=5.0.0->otter_grader->-r ./requirements.txt (line 1)) (0.27.1)

Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-packages (from jupyter-client>=6.1.12->ipykernel->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.4)

Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.7->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (4.4.0)

Requirement already satisfied: jupyter-events>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.12.0)

Requirement already satisfied: jupyter-server-terminals>=0.4.4 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (0.5.3)

Requirement already satisfied: overrides>=5.0 in /usr/local/lib/python3.12/dist-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (7.7.0)

Requirement already satisfied: websocket-client>=1.7 in

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/usr/local/lib/python3.12/dist-packages (from jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(1.8.0)
Requirement already satisfied: argon2-cffi-bindings in
/usr/local/lib/python3.12/dist-packages (from
argon2-cffi->notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9)) (25.1.0)
Requirement already satisfied: babel>=2.10 in /usr/local/lib/python3.12/dist-
packages (from jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (2.17.0)
Collecting json5>=0.9.0 (from jupyterlab-
server<3,>=2.27.1->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
  Downloading json5-0.12.1-py3-none-any.whl.metadata (36 kB)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-
packages (from markdown-it-py>=1.0->jupyter-text<2.0.0,>=1.16.4->otter_grader->-r
./requirements.txt (line 1)) (0.1.2)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from
pexpect>4.3->ipython->otter_grader->-r ./requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->otter_grader->-r
./requirements.txt (line 1)) (0.2.13)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,<3,>=2->python-on-
whales<1.0.0,>=0.72.0->otter_grader->-r ./requirements.txt (line 1)) (0.7.0)
Requirement already satisfied: pydantic-core==2.33.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,<3,>=2->python-on-
whales<1.0.0,>=0.72.0->otter_grader->-r ./requirements.txt (line 1)) (2.33.2)
Requirement already satisfied: typing-inspection>=0.4.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,<3,>=2->python-on-
whales<1.0.0,>=0.72.0->otter_grader->-r ./requirements.txt (line 1)) (0.4.1)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-
packages (from beautifulsoup4->nbconvert>=6.0.0->nbconvert[webpdf]>=6.0.0;
sys_platform != "emscripten" and sys_platform != "wasi"->otter_grader->-r
./requirements.txt (line 1)) (2.8)
Collecting pyee<14,>=13 (from playwright->nbconvert[webpdf]>=6.0.0; sys_platform
!= "emscripten" and sys_platform != "wasi"->otter_grader->-r ./requirements.txt
(line 1))
  Downloading pyee-13.0.0-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: greenlet<4.0.0,>=3.1.1 in
/usr/local/lib/python3.12/dist-packages (from
playwright->nbconvert[webpdf]>=6.0.0; sys_platform != "emscripten" and
sys_platform != "wasi"->otter_grader->-r ./requirements.txt (line 1)) (3.2.4)
Requirement already satisfied: sphinxcontrib-applehelp>=1.0.7 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-devhelp>=1.0.6 in
/usr/local/lib/python3.12/dist-packages (from

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sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-htmlhelp>=2.0.6 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.1.0)
Requirement already satisfied: sphinxcontrib-jsmath>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (1.0.1)
Requirement already satisfied: sphinxcontrib-qthelp>=1.0.6 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: sphinxcontrib-serializinghtml>=1.1.9 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (2.0.0)
Requirement already satisfied: snowballstemmer>=2.2 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (3.0.1)
Requirement already satisfied: alabaster>=0.7.14 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (1.0.0)
Requirement already satisfied: imagesize>=1.3 in /usr/local/lib/python3.12/dist-
packages (from sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line
1)) (1.4.1)
Requirement already satisfied: roman-numerals-py>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from
sphinx->fica>=0.4.1->otter_grader->-r ./requirements.txt (line 1)) (3.1.0)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-
packages (from anyio->httpx<1,>=0.25.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.3.1)
Requirement already satisfied: python-json-logger>=2.0.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(3.3.0)
Requirement already satisfied: rfc3339-validator in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in
/usr/local/lib/python3.12/dist-packages (from jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(0.1.1)
Requirement already satisfied: cffi>=1.0.1 in /usr/local/lib/python3.12/dist-
packages (from argon2-cffi-bindings->argon2-cffi->notebook->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (2.0.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-
packages (from cffi>=1.0.1->argon2-cffi-
bindings->argon2-cffi->notebook->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(2.23)
Requirement already satisfied: fqdn in /usr/local/lib/python3.12/dist-packages

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(from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(1.5.1)
Requirement already satisfied: isoduration in /usr/local/lib/python3.12/dist-
packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (20.11.0)
Requirement already satisfied: jsonpointer>1.1.3 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(3.0.0)
Requirement already satisfied: rfc3987-syntax>=1.1.0 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(1.1.0)
Requirement already satisfied: uri-template in /usr/local/lib/python3.12/dist-
packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in
/usr/local/lib/python3.12/dist-packages (from jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(24.11.1)
Requirement already satisfied: lark>=1.2.2 in /usr/local/lib/python3.12/dist-
packages (from rfc3987-syntax>=1.1.0->jsonschema[format-
nongpl]>=4.18.0->jupyter-events>=0.9.0->jupyter-
server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r ./requirements.txt (line 9))
(1.2.2)
Requirement already satisfied: arrow>=0.15.0 in /usr/local/lib/python3.12/dist-
packages (from isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (1.3.0)
Requirement already satisfied: types-python-dateutil>=2.8.10 in
/usr/local/lib/python3.12/dist-packages (from
arrow>=0.15.0->isoduration->jsonschema[format-nongpl]>=4.18.0->jupyter-
events>=0.9.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter>=1.0.0->-r
./requirements.txt (line 9)) (2.9.0.20250822)
Downloading jupyter-1.1.1-py2.py3-none-any.whl (2.7 kB)
Downloading fica-0.4.1-py3-none-any.whl (13 kB)
Downloading ipylab-1.1.0-py3-none-any.whl (101 kB)
101.6/101.6 kB
5.7 MB/s eta 0:00:00
Downloading ipywidgets-8.1.7-py3-none-any.whl (139 kB)
139.8/139.8 kB
12.0 MB/s eta 0:00:00

```

```

Downloading python_on_whales-0.78.0-py3-none-any.whl (118 kB)
118.1/118.1 kB
9.9 MB/s eta 0:00:00
Downloading jupyterlab-4.4.7-py3-none-any.whl (12.3 MB)
12.3/12.3 MB
82.9 MB/s eta 0:00:00
Downloading async_lru-2.0.5-py3-none-any.whl (6.1 kB)
Downloading comm-0.2.3-py3-none-any.whl (7.3 kB)
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
64.9 MB/s eta 0:00:00
Downloading jupyter_lsp-2.3.0-py3-none-any.whl (76 kB)
76.7/76.7 kB
5.9 MB/s eta 0:00:00
Downloading jupyterlab_server-2.27.3-py3-none-any.whl (59 kB)
59.7/59.7 kB
5.0 MB/s eta 0:00:00
Downloading widgetsnbextension-4.0.14-py3-none-any.whl (2.2 MB)
2.2/2.2 MB
81.9 MB/s eta 0:00:00
Downloading playwright-1.55.0-py3-none-manylinux1_x86_64.whl (45.9 MB)
45.9/45.9 MB
22.6 MB/s eta 0:00:00
Downloading json5-0.12.1-py3-none-any.whl (36 kB)
Downloading pyee-13.0.0-py3-none-any.whl (15 kB)
Building wheels for collected packages: otter_grader
  Building wheel for otter_grader (pyproject.toml) ... done
  Created wheel for otter_grader: filename=otter_grader-6.1.3-py3-none-any.whl
  size=142409
  sha256=9635982965081b549d1a1cef65a95dc3d2fec63425cc7d1eaff32d612bea158d
  Stored in directory: /tmp/pip-ephem-wheel-cache-
  jhz2chum/wheels/6c/c4/a1/ed40bf2584b0c3446e20b8a0aea721771c46fe292cffe8e50b
Successfully built otter_grader
Installing collected packages: widgetsnbextension, pyee, json5, jedi, comm,
async-lru, playwright, python-on-whales, ipywidgets, fica, ipylab, otter_grader,
jupyterlab-server, jupyter-lsp, jupyterlab, jupyter
  Attempting uninstall: widgetsnbextension
    Found existing installation: widgetsnbextension 3.6.10
    Uninstalling widgetsnbextension-3.6.10:
      Successfully uninstalled widgetsnbextension-3.6.10
  Attempting uninstall: ipywidgets
    Found existing installation: ipywidgets 7.7.1
    Uninstalling ipywidgets-7.7.1:
      Successfully uninstalled ipywidgets-7.7.1
Successfully installed async-lru-2.0.5 comm-0.2.3 fica-0.4.1 ipylab-1.1.0
ipywidgets-8.1.7 jedi-0.19.2 json5-0.12.1 jupyter-1.1.1 jupyter-lsp-2.3.0
jupyterlab-4.4.7 jupyterlab-server-2.27.3 otter_grader-6.1.3 playwright-1.55.0
pyee-13.0.0 python-on-whales-0.78.0 widgetsnbextension-4.0.14

```

```

Collecting kaleido
  Downloading kaleido-1.1.0-py3-none-any.whl.metadata (5.6 kB)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
(5.24.1)
Collecting plotly
  Downloading plotly-6.3.0-py3-none-any.whl.metadata (8.5 kB)
Collecting choreographer>=1.0.10 (from kaleido)
  Downloading choreographer-1.1.1-py3-none-any.whl.metadata (6.8 kB)
Collecting logistro>=1.0.8 (from kaleido)
  Downloading logistro-1.1.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: orjson>=3.10.15 in
/usr/local/lib/python3.12/dist-packages (from kaleido) (3.11.3)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from kaleido) (25.0)
Collecting pytest-timeout>=2.4.0 (from kaleido)
  Downloading pytest_timeout-2.4.0-py3-none-any.whl.metadata (20 kB)
Requirement already satisfied: narwhals>=1.15.1 in
/usr/local/lib/python3.12/dist-packages (from plotly) (2.5.0)
Requirement already satisfied: simplejson>=3.19.3 in
/usr/local/lib/python3.12/dist-packages (from choreographer>=1.0.10->kaleido)
(3.20.1)
Requirement already satisfied: pytest>=7.0.0 in /usr/local/lib/python3.12/dist-
packages (from pytest-timeout>=2.4.0->kaleido) (8.4.2)
Requirement already satisfied: iniconfig>=1 in /usr/local/lib/python3.12/dist-
packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (2.1.0)
Requirement already satisfied: pluggy<2,>=1.5 in /usr/local/lib/python3.12/dist-
packages (from pytest>=7.0.0->pytest-timeout>=2.4.0->kaleido) (1.6.0)
Requirement already satisfied: pygments>=2.7.2 in
/usr/local/lib/python3.12/dist-packages (from pytest>=7.0.0->pytest-
timeout>=2.4.0->kaleido) (2.19.2)
Downloading kaleido-1.1.0-py3-none-any.whl (66 kB)
      66.3/66.3 kB
3.4 MB/s eta 0:00:00
Downloading plotly-6.3.0-py3-none-any.whl (9.8 MB)
      9.8/9.8 MB
78.2 MB/s eta 0:00:00
Downloading choreographer-1.1.1-py3-none-any.whl (52 kB)
      52.3/52.3 kB
4.4 MB/s eta 0:00:00
Downloading logistro-1.1.0-py3-none-any.whl (7.9 kB)
Downloading pytest_timeout-2.4.0-py3-none-any.whl (14 kB)
Installing collected packages: plotly, logistro, pytest-timeout, choreographer,
kaleido
  Attempting uninstall: plotly
    Found existing installation: plotly 5.24.1
    Uninstalling plotly-5.24.1:
      Successfully uninstalled plotly-5.24.1
Successfully installed choreographer-1.1.1 kaleido-1.1.0 logistro-1.1.0

```

plotly-6.3.0 pytest-timeout-2.4.0

```
[2]: # Initialize Otter
import otter
grader = otter.Notebook("fashion_pt_2.ipynb")
```

Homework 1.2 – AGI, Everywhere, All at Once

Welcome to Homework 1.2! In this assignment, you will build on the foundational skills you developed in part 1. You will gain hands-on experience with building and evaluating machine learning models, debugging performance issues, and exploring strategies to improve model accuracy. By the end of this assignment, you will have a deeper understanding of how to work with classifiers and analyze their performance.

0.1 Due Date: Friday, September 26, 11:59 PM

This assignment is due on **Friday, September 26, at 11:59 PM**. You must submit your work to Gradescope by this deadline. Please refer to the syllabus for the [Slip Day policy](#). No late submissions will be accepted beyond the details outlined in the Slip Day policy.

0.1.1 Submission Tips:

- **Plan ahead:** We strongly encourage you to submit your work several hours before the deadline. This will give you ample time to address any submission issues.
 - **Reach out for help early:** If you encounter difficulties, contact course staff well before the deadline. While we are happy to assist with submission issues, we cannot guarantee responses to last-minute requests.
-

0.2 Assignment Overview

This notebook contains a series of tasks designed to help you practice and apply key concepts in machine learning. You will complete all the TODOs in the notebook, which include both coding and written response questions. Some tasks are open-ended, which allows you to explore and experiment with different approaches.

0.2.1 Key Learning Objectives:

1. Build and evaluate machine learning classifiers.
 2. Debug and analyze model performance.
 3. Explore techniques to improve accuracy.
 4. Gain experience with tools like `numpy`, `pandas`, `plotly`, and `scikit-learn`.
-

0.2.2 Grading Breakdown

Question	Manual Grading?	Points
5a	No	2
5b	No	4
5c	No	1
5d	Yes	2
6a	No	3
6b	Yes	2
6c	No	1
6d	Yes	1
7a	No	3
7b	Yes	3
8a	Yes	3
8b	No	3
8c	No	5
8d	Yes	1
9a	No	5
Total		39

Note: “Manual” questions are written response questions that will be graded manually by the course staff. All other questions will be graded automatically by the autograder.

0.2.3 Instructions:

1. Carefully read each question and its requirements.
2. Complete all TODOs in the notebook. You may add extra lines of code if needed to implement your solution.
3. For manual questions, provide clear and concise written responses.
4. Test your code thoroughly to ensure it meets the requirements.

Good luck!

```
[3]: import numpy as np
import pandas as pd
import plotly.express as px
import torchvision
import os
from scipy.ndimage import rotate
import random
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
import plotly.graph_objects as go
from sklearn.neural_network import MLPRegressor
from sklearn.metrics import confusion_matrix
from sklearn.linear_model import LinearRegression
from sklearn.neural_network import MLPClassifier
```

```

from sklearn.preprocessing import StandardScaler
import joblib # For saving/loading the model

pd.options.plotting.backend = "plotly"

```

0.2.4 IMPORTANT:

- Do not change the random seed values!!!
- Before you submit your notebook, remember to set `save_models=True` and `load_models=True`. This saves your final models which we will use for the autograder. Set these to false if you are still tweaking your model setup. We have provided code for saving models - **do not change these file names!!**
- When uploading your notebook, make sure to include your model file `classifier.joblib` in your submission

```

[4]: # Set random seeds for reproducible results
SEED = 42
np.random.seed(SEED)
random.seed(SEED)

# IMPORTANT: set save_models to True to save trained models. YOU NEED TO DO
↳ THIS FOR THE AUTOGRADER TO WORK.
save_models = True
load_saved_models = True # After training, you can set this to True to load the
↳ saved models and not have to re-train them.
IS_GRADING_ENV = os.getenv("IS_GRADING_ENV") == "true"

```

```

[5]: # Importing helper function to show images
def show_images(images, max_images=40, ncols=5, labels = None, reshape=False):
    """Visualize a subset of images from the dataset.
    Args:
        images (np.ndarray or list): Array of images to visualize [img,row,col].
        max_images (int): Maximum number of images to display.
        ncols (int): Number of columns in the grid.
        labels (np.ndarray, optional): Labels for the images, used for facet
    ↳ titles.
    Returns:
        plotly.graph_objects.Figure: A Plotly figure object containing the
    ↳ images.
    """
    if isinstance(images, list):
        images = np.stack(images)
    n = min(images.shape[0], max_images) # number of images to show
    px_height = 220 # height of each image in pixels
    if reshape:
        images = images.reshape(images.shape[0], 28, 28)
    fig = px.imshow(images[:n, :, :], color_continuous_scale='gray_r',

```

```

        facet_col = 0, facet_col_wrap=ncols,
        height = px_height * int(np.ceil(n/ncols)))
fig.update_layout(coloraxis_showscale=False)
fig.update_xaxes(showticklabels=False, showgrid=False)
fig.update_yaxes(showticklabels=False, showgrid=False)
if labels is not None:
    # Extract the facet number and replace with the label.
    fig.for_each_annotation(lambda a: a.update(text=labels[int(a.text.
↪split("=")[-1]))))
    return fig

```

0.3 Load the Fashion-MNIST Dataset

Let's reload the Fashion-MNIST dataset. As a reminder, this dataset contains 70k images (60k training, 10k testing) of 28x28 grayscale images of clothing divided into the following 10 classes: 1. **T-shirt/top** 2. **Trouser** 3. **Pullover** 4. **Dress** 5. **Coat** 6. **Sandal** 7. **Shirt** 8. **Sneaker** 9. **Bag** 10. **Ankle boot**

0.3.1 Dataset Structure:

- **Images:** Each image is represented as a 28x28 grayscale matrix, where each value corresponds to the pixel intensity.
- **Targets:** Each image is associated with a label (0-9), which maps to one of the 10 classes listed above.

0.3.2 Data Preparation:

In the code cell below, we: 1. Load the Fashion-MNIST training dataset using `torchvision.datasets.FashionMNIST`. 2. Extract the images and targets into NumPy arrays for easier manipulation. 3. Create a pandas DataFrame (df) to store the images and their corresponding labels.

```

[6]: # Load the FashionMNIST training dataset
train_data = torchvision.datasets.FashionMNIST(root='./data', train=True,
↪download=True)
images = train_data.data.numpy().astype(float) # Convert images to numpy array
targets = train_data.targets.numpy() # Extract labels
class_dict = {i:class_name for i,class_name in enumerate(train_data.classes)}
labels = np.array([class_dict[t] for t in targets]) # Map labels to class names
n = len(images) # Total samples
class_names = list(class_dict.values())
print(f"Loaded FashionMNIST with {n} samples. Classes: {class_dict}")
print("Classes: {}".format(class_dict))
print("Image shape: {}".format(images[0].shape))
print("Image dtype: {}".format(images[0].dtype))

# Creating a DataFrame with the images and labels

```

```
df = pd.DataFrame({"image": images.tolist(), "label": labels})
# Cast image as numpy array
df['image'] = df['image'].apply(lambda x: np.array(x).reshape(-1))
```

Loaded FashionMNIST with 60000 samples. Classes: {0: 'T-shirt/top', 1: 'Trouser', 2: 'Pullover', 3: 'Dress', 4: 'Coat', 5: 'Sandal', 6: 'Shirt', 7: 'Sneaker', 8: 'Bag', 9: 'Ankle boot'}

Classes: {0: 'T-shirt/top', 1: 'Trouser', 2: 'Pullover', 3: 'Dress', 4: 'Coat', 5: 'Sandal', 6: 'Shirt', 7: 'Sneaker', 8: 'Bag', 9: 'Ankle boot'}

Image shape: (28, 28)

Image dtype: float64

0.3.3 (Training) Fitting a Classifier

Just like in part 1, let's train a 2-layer MLP classifier on the original data and see how it performs.

Task: Copy and paste your code from part 1 (problem 3c).

Below is the reminder about the process for training any model that will follow in the next question:

0.3.4 Steps:

1. **Split the Data:** Use [scikit-learn train_test_split](#) to divide the dataset into training and testing subsets.
2. **Standardize the Features:** Use [scikit-learn StandardScaler](#) to scale the pixel values of the images.
3. **Train the Classifier:** Use [scikit-learn MLPClassifier](#) to train a neural network on the training data.
4. **Evaluate the Model:** Measure the model's accuracy on both the training and testing datasets.

Why Train-Test Split? We split the dataset into training and testing subsets to evaluate the model's performance on unseen data. This helps us evaluate the model's generalization capability.

Why Standardize the Data? Standardization scales the input features to have zero mean and unit variance. This helps the MLP converge faster during training and improves the stability of the optimization process.

Useful docs: * [scikit-learn train_test_split](#) * [scikit-learn MLPClassifier](#) * [scikit-learn Standard Scaler](#)

```
[7]: # Split the dataset into training and testing sets (80% train, 20% test)
train_df, test_df = train_test_split(df, test_size=0.2, random_state=SEED)

# Extract image data and labels for training and testing
X_train = np.stack(train_df['image'].to_numpy()) # Training images
X_test = np.stack(test_df['image'].to_numpy())   # Testing images
y_train = train_df['label'].to_numpy()          # Training labels
y_test = test_df['label'].to_numpy()             # Testing labels

# Print the shapes of the training and testing datasets
```

```
print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
```

X_train shape: (48000, 784)

X_test shape: (12000, 784)

```
[8]: from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler
import joblib # for saving the model

if (load_saved_models or IS_GRADING_ENV) and os.path.
    exists('mlp_fashionmnist_model.joblib'):
    model = joblib.load("mlp_fashionmnist_model.joblib")
    scaler = StandardScaler()
    X_train_sc = scaler.fit_transform(X_train)
    X_test_sc = scaler.transform(X_test)
else:
    # TODO: paste your training code from part 1 (problem 3c) here setting
    random_state=SEED
    # flatten features into 1D arrays
    X_train = np.stack(train_df['image'].values)
    y_train = train_df['label'].values
    X_test = np.stack(test_df['image'].values)
    y_test = test_df['label'].values

    print(f"X_train shape: {X_train.shape}\t y_train shape: {y_train.shape}")
    print(f"X_test shape: {X_test.shape}\t y_test shape: {y_test.shape}")

    # TODO: Train the model using the scaled training data and plot the loss
    curve (remember to normalize your data!)
    scaler = StandardScaler()
    scaler.fit(X_train)
    X_train_sc = scaler.transform(X_train)
    # fitting helps to "shift" data such that origin is aligned to training set.
    # test data and eval should undergo same shift for model to predict
    accurately
    # so don't fit on every new set of data
    #scaler.fit(X_test)
    X_test_sc = scaler.transform(X_test)

    # NOTE: Your model must be named `model`
    model = MLPClassifier(hidden_layer_sizes=(100, 50), max_iter=100, tol=1e-3,
    random_state=SEED)

    model.fit(X=X_train_sc, y=y_train)
    loss_curve = model.loss_curve_
```

```

    loss_df = pd.DataFrame({'epoch': range(len(loss_curve)), 'loss': loss_curve})
    ↪loss_curve})
if save_models:
    # Save the trained model and scaler to disk
    joblib.dump(model, "mlp_fashionmnist_model.joblib")
    print("Model and scaler saved to disk.")

train_df['predicted_label'] = model.predict(X_train_sc)
test_df['predicted_label'] = model.predict(X_test_sc)

train_df['correct'] = train_df['predicted_label'] == train_df['label']
test_df['correct'] = test_df['predicted_label'] == test_df['label']

print(f"Training accuracy: {train_df['correct'].mean():.3f}")
print(f"Test accuracy: {test_df['correct'].mean():.3f}")

```

Model and scaler saved to disk.

Training accuracy: 0.993

Test accuracy: 0.884

0.4 Problem 5: Regression Analysis

In this section, we will switch from classification to regression. The goal is to predict the price of clothing items based on their image features. Follow the steps below to load the price data, train a regression model, and evaluate its performance.

0.4.1 Problem 5a: Training a Linear Regression Model

Task: Use pandas and scikit-learn to train a linear regression model to predict prices:

1. Join the DataFrames:

- Combine the price data with the original `train_df` and `test_df` using the `join` method.
- Store the resulting training data in `prices_train` and the testing data in `prices_test`.

Hint: Think about which column can serve as the key for joining the `prices` DataFrame with the original `train_df` and `test_df`.

2. Train the Model:

- Use `LinearRegression` from `sklearn.linear_model` to train a regression model.
- Ensure that the model predicts only positive prices.

3. Store Predictions:

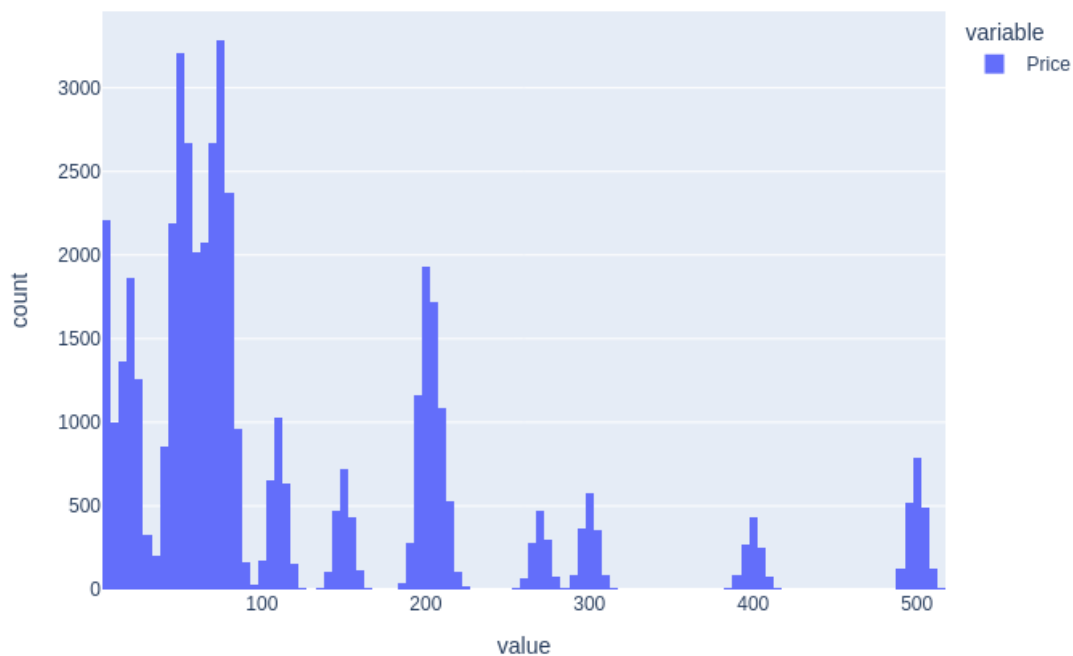
- Add the predicted prices to the `prices_test` DataFrame.

Important for grading: Name your model `price_model`.

Expected Output: The `prices_train` and `prices_test` DataFrames should look like this:

key_0	image	label	predicted_label	correct	Price
48572	[0.0, 0.0, ...]	Sneaker	Sneaker	True	49.86
38696	[0.0, 0.0, ...]	Dress	Dress	True	14.56

```
[9]: prices = pd.read_csv("../data/FashionMNIST_prices.csv")
prices_train_portion, prices_test_portion = train_test_split(prices,
↳test_size=0.2, random_state=SEED)
# TODO: Join the original `train_df` and `test_df` with the `prices` DataFrame
prices_train = train_df.join(prices_train_portion, on=train_df.index)
prices_test = test_df.join(prices_test_portion, on=test_df.index)
prices_train['Price'].hist().show()
prices_train.head()
prices_train.shape
```



[9]: (48000, 6)

```
[10]: # TODO: Fit a linear regression model (price_model) on the training data with
↳prices as the target variable
if (load_saved_models or IS_GRADING_ENV) and os.path.exists('price_model.
↳joblib'):
    price_model = joblib.load('price_model.joblib')
```

```

price_model = joblib.load('price_model.joblib')
print("Model loaded from price_model.joblib")
else:
    price_model = LinearRegression()
    price_model.fit(X=X_train, y=prices_train['Price'])
    print("Model trained on training data")

# TODO: Predict the prices of the test data and add them to the test_df
prices_test['price_prediction'] = price_model.predict(X_test)
display(prices_test.head())
prices_test['price_prediction'] = np.clip(prices_test['price_prediction'], 0,
↳None)

if save_models:
    joblib.dump(price_model, 'price_model.joblib')
    print("Model saved to price_model.joblib")

```

Model loaded from price_model.joblib

	key_0		image	label \
12628	12628	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
37730	37730	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
39991	39991	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Sneaker
8525	8525	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, ...		Coat
8279	8279	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag

	predicted_label	correct	Price	price_prediction
12628	Bag	True	405.81	332.356362
37730	Bag	True	202.36	243.933962
39991	Sneaker	True	39.15	14.335877
8525	Coat	True	68.28	168.062470
8279	Bag	True	395.15	406.643727

Model saved to price_model.joblib

```
[11]: grader.check("q5a")
```

[11]: q5a results: All test cases passed!

0.4.2 Problem 5b: Evaluate Linear Regression Performance

Task: Use NumPy operations to evaluate the performance of the linear regression model:

1. Calculate the **Root Mean Squared Error (RMSE)** to measure the average magnitude of prediction errors.
2. Calculate the **Mean Absolute Error (MAE)** to measure the average absolute difference between predicted and actual values.

3. Calculate the **R-squared score (R^2)** to assess how well the model explains the variance in the data.

```
[12]: def compute_rmse(y_true, y_pred):
    return np.sqrt(mean_squared_error(y_true, y_pred))

def compute_mae(y_true, y_pred):
    return np.mean(np.abs(y_true-y_pred))

def compute_r2(y_true, y_pred):
    """Compute R-squared score"""
    y_mean = np.mean(y_true)
    ss_tot = np.sum((y_true - y_mean) ** 2)
    ss_res = np.sum((y_true - y_pred) ** 2)
    r2 = 1 - (ss_res / ss_tot)
    return r2

def print_metrics(y_true, y_pred, dataset=""):
    """Print all regression metrics for a dataset"""
    rmse = compute_rmse(y_true, y_pred)
    mae = compute_mae(y_true, y_pred)
    r2 = compute_r2(y_true, y_pred)

    print(f"=== {dataset} Metrics ===")
    print(f"RMSE: {rmse:.2f}")
    print(f"MAE: {mae:.2f}")
    print(f"R2: {r2:.3f}")
    return rmse, mae, r2

# Calculate predictions
y_train_pred = price_model.predict(X_train_sc)
y_test_pred = price_model.predict(X_test_sc)
y_test_pred = np.clip(y_test_pred, 0, None)

# Compute and print metrics
train_rmse, train_mae, train_r2 = print_metrics(prices_train['Price'],
    ↪ y_train_pred, "Training")
test_rmse, test_mae, test_r2 = print_metrics(prices_test['Price'], y_test_pred,
    ↪ "Test")

# Visualize predictions vs actual
fig = px.scatter(
    x=prices_test['Price'],
    y=y_test_pred,
    title='Predicted vs Actual Prices',
```

```

    labels={'x': 'Actual Price', 'y': 'Predicted Price'}
)
fig.add_trace(px.line(x=[prices_test['Price'].min(), prices_test['Price'].
    ↪max()],
                    y=[prices_test['Price'].min(), prices_test['Price'].
    ↪max()])).data[0])
fig.show()

```

=== Training Metrics ===

RMSE: 140.33

MAE: 86.01

R^2 : -0.396

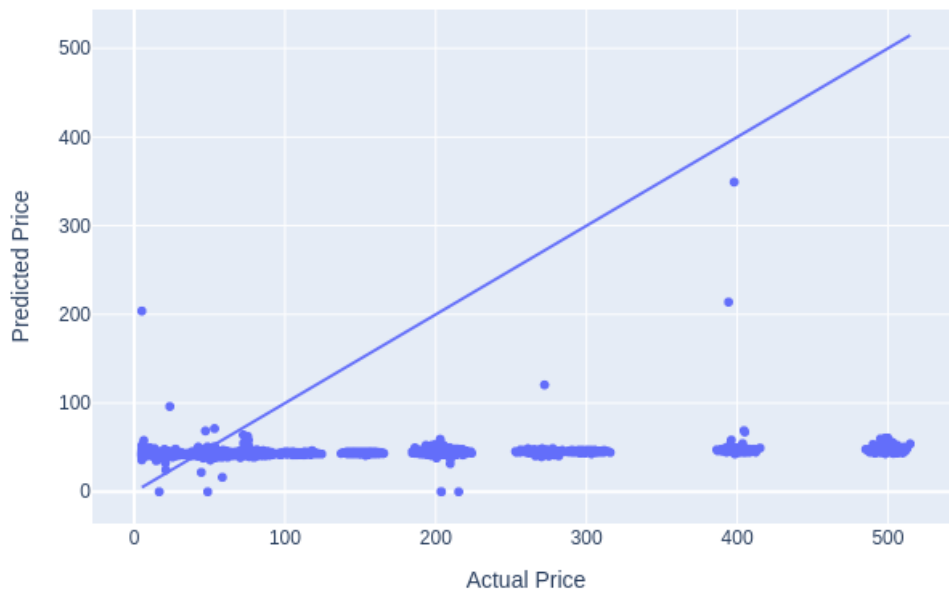
=== Test Metrics ===

RMSE: 138.06

MAE: 84.70

R^2 : -0.397

Predicted vs Actual Prices



```
[13]: grader.check("q5b")
```

[13]: q5b results: All test cases passed!

0.4.3 Problem 5c: MAE, MSE, and R-Squared

Question: What are the advantages of using MAE over MSE to measure a model's performance? What are the advantages of using MSE over MAE to measure a model's performance?

Errors in prediction for a Linear Regression Model may be positive or negative depending on the difference between the prediction and the target value.

MAE: Mean Absolute Error Measures error in the same units as the prediction for easier interpretability. Doesn't overpenalise large errors unlike MSE.

MSE: Mean Squared Error Using MSE as a loss function makes the loss function differentiable. This facilitates gradient descent as gradients can be calculated easily and model parameters can be updated. MSE also overpenalises large errors (outliers) which might be a desirable trait depending on the dataset and the Linear Regression's task.

Question: What does R-squared measure? Why is it significant?

R-squared measures the proportion of the variance of the target variable captured by the model. The higher the value of R-squared, the better the model is fit to the data. It is significant because it can be used as a metric to understand how accurate the model's predictions are.

0.4.4 Problem 5d: Relationship Between Classification and Regression Errors

Task: Analyze the relationship between classification and regression errors: 1. For each test sample, calculate the price prediction error and the relative price prediction error: - Add the following columns to `prices_test`: - `price_error`: The absolute difference between the predicted price and the actual price. - `price_error_relative`: The relative error, calculated as the absolute error divided by the actual price. 2. Compute and print: - `misclassified_price_error`: The average relative price error for misclassified images. - `correctly_classified_price_error`: The average relative price error for correctly classified images.

Hints: - Use `np.abs` to calculate the absolute value of the price error. - Divide the price error by the actual price to compute the relative price error.

```
[14]: # TODO: Compare how accurate the model is at classifying the images vs how
      ↪ accurate it is at predicting the price of the images
      # TODO: Calculate the average relative price error for correctly classified vs.
      ↪ misclassified images
      display(prices_test)
      prices_test['price_error'] = np.abs(prices_test['Price'] -
      ↪ prices_test['price_prediction'])
      prices_test['price_error_relative'] = prices_test['price_error'] /
      ↪ prices_test['Price']
      misclassified_price_error = prices_test[prices_test['correct'] ==
      ↪ False]['price_error_relative'].mean()
      correctly_classified_price_error = prices_test[prices_test['correct'] ==
      ↪ True]['price_error_relative'].mean()

      print(f"\nPrice prediction performance:")
```

```
print(f"Misclassified images - Avg relative price error:␣
↪{misclassified_price_error:.3f}")
print(f"Correctly classified images - Avg relative price error:␣
↪{correctly_classified_price_error:.3f}")
```

	key_0		image	label \
12628	12628	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
37730	37730	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
39991	39991	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Sneaker
8525	8525	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, ...		Coat
8279	8279	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
...	...		...	...
49914	49914	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Sneaker
34428	34428	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Shirt
45810	45810	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
30357	30357	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...		Bag
36568	36568	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 63.0, 62.0...		Trouser

	predicted_label	correct	Price	price_prediction
12628	Bag	True	405.81	332.356362
37730	Bag	True	202.36	243.933962
39991	Sneaker	True	39.15	14.335877
8525	Coat	True	68.28	168.062470
8279	Bag	True	395.15	406.643727
...	...	...	...	...
49914	Sandal	False	41.42	51.156599
34428	Shirt	True	8.58	27.335636
45810	Bag	True	204.73	95.023217
30357	Bag	True	395.34	383.948469
36568	Shirt	False	77.93	126.708765

[12000 rows x 7 columns]

Price prediction performance:

Misclassified images - Avg relative price error: 1.472

Correctly classified images - Avg relative price error: 0.767

```
[15]: grader.check("q5d")
```

```
[15]: q5d results: All test cases passed!
```

0.4.5 Problem 5e: Relationship Between Classification and Price Prediction Errors

Question 1. Is there a relationship between classification errors and price prediction errors? If so, what is the nature of this relationship?

2. Why might classification errors correlate with price prediction errors? Provide possible ex-

planations.

3. Why might classification errors and price prediction errors be unrelated? Discuss potential reasons.
 1. It seems like the model predicts prices more accurately for correctly classified images evidenced by the lower average relative price error compared to misclassified images. There is a positive correlation between accuracy of classification and accuracy of price prediction.
 2. Since both models receive the same input of images (or pixels of the image arranged in a NumPy array), the articles of clothing that the classification model misclassifies might contain information which the model struggles to learn from or characterise. This may also impact the Linear Regression model's ability to predict the price of the same article of clothing due to the inability of the model to understand the article of clothing.
 3. They might be unrelated as the input to the classifier and regression are solving different problems. The output of the classification model is a label categorising the article of clothing. On the other hand, the Linear Regression model is trying to predict the price of the clothing. Even for clothes which have been correctly identified, the Linear Regression model has to account for other characteristics of the clothing to accurately predict the price of the clothing as even within the same category of clothing the price may differ greatly

0.5 Problem 6: Working with a New Test Set

In this section, we will load a **super secret** test set. Your goal is to maximize the model's accuracy on this dataset.

0.5.1 Problem 6a: Inference Using model

Task: Use your `model` (the MLP classifier model trained earlier in this notebook) to predict the labels of the secret test set.

1. Create a `DataFrame` named `test_secret_df` to store the secret test set data.
2. Use the `model` to predict the image labels for the secret test set.

Note: Remember to use `scaler.transform` to preprocess the secret test set before making predictions. This ensures the data is scaled consistently with the training data. Learn more about `scaler.transform` [here](#).

```
[16]: X_test_secret = np.load("../data/secret_test_set/X_test.npy")
      y_test_secret = np.load("../data/secret_test_set/y_test.npy")

      X_test_secret_sc = scaler.transform(X_test_secret)

      print(X_test_secret.shape)
      print(y_test_secret.shape)
```

```
(2000, 784)
(2000,)
```

```
[17]: # TODO: Create a dataframe with the secret test set and use the model to
      ↪predict the labels
test_secret_df = pd.DataFrame({"image": X_test_secret.tolist(), "label":
      ↪y_test_secret})
test_secret_df['predicted_label'] = model.predict(X_test_secret_sc)

# TODO: Use the model to predict the labels and calculate the accuracy
test_secret_df['correct'] = test_secret_df['predicted_label'] ==
      ↪test_secret_df['label']

print(f"Test accuracy: {test_df['correct'].mean():.3f}")
print(f"Secret Test accuracy: {test_secret_df['correct'].mean():.3f}")
```

Test accuracy: 0.884
Secret Test accuracy: 0.205

```
[18]: grader.check("q6a")
```

[18]: q6a results: All test cases passed!

0.5.2 Problem 6b: Class-wise Accuracy Comparison

The test accuracy is significantly lower than the training accuracy. To investigate this, let's examine the class-wise accuracies.

Task: 1. Plot the class-wise accuracy for both the original test set and the secret test set as a bar plot, using different hues to distinguish between the two datasets.

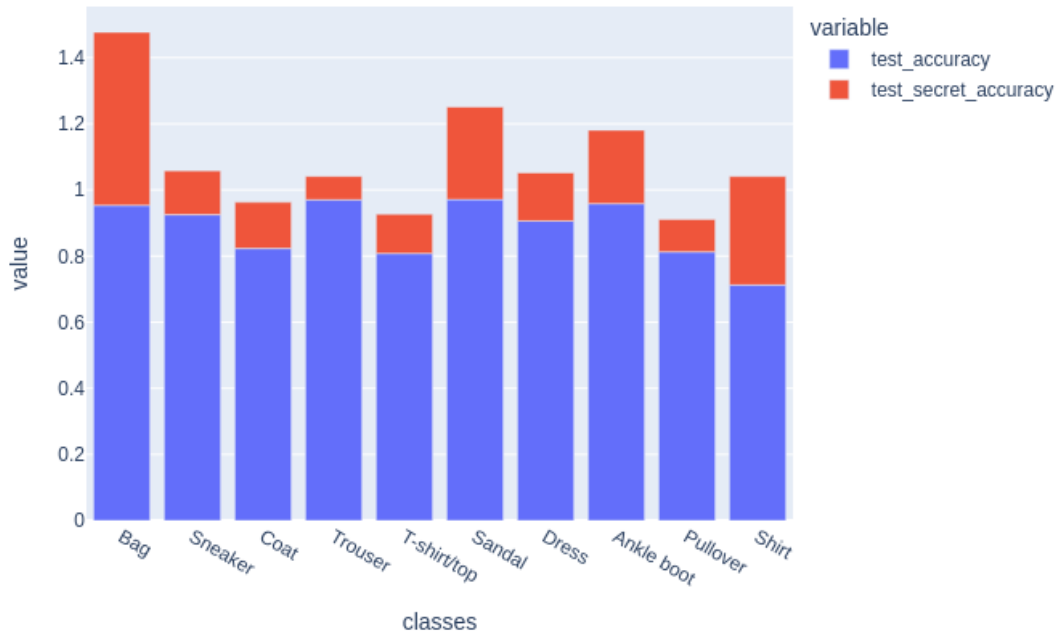
Hint: You can use [Pandas's plotting functionality](#) to create the bar plot.

Expected Output: A bar plot with two bars for each class—one representing the model's accuracy on the original test set and the other representing the accuracy on the secret test set.

```
[19]: # TODO: Make a two-colored bar plot of the model's accuracy on the test set and
      ↪the secret test set
plotted_data = pd.DataFrame({'classes': test_df['label'].unique(),
      ↪'test_accuracy': np.zeros(len(test_df['label'].unique())),
      ↪'test_secret_accuracy': np.zeros(len(test_df['label'].unique()))})
# for each class, check accuracy of test set and secret test set
tmp = test_df.groupby('label')
for x,y in tmp:
    plotted_data.loc[plotted_data['classes'] == x, 'test_accuracy'] =
      ↪y['correct'].mean()
tmp2 = test_secret_df.groupby('label')
for x,y in tmp2:
    plotted_data.loc[plotted_data['classes'] == x, 'test_secret_accuracy'] =
      ↪y['correct'].mean()
```

```
plt = plotted_data.plot(kind='bar', x='classes',  
    y=['test_accuracy', 'test_secret_accuracy'], title='Class-Wise Model Accuracy  
    on Test and Secret Test Sets')  
plt.show()
```

Class-Wise Model Accuracy on Test and Secret Test Sets



0.5.3 Problem 6c: Confusion Matrix

To better understand why the model's accuracy is low on the super secret test set, let's analyze the confusion matrix.

Task: 1. Generate a confusion matrix comparing the true labels (`y_test_secret`) with the model's predictions on the secret test set.

Hint: Use `confusion_matrix` from `scikit-learn`.

Expected Output: A 10 x 10 confusion matrix with the FASHION class labels on both axes.

```
[20]: # TODO: plot a confusion matrix for the secret test set  
conf_matrix = confusion_matrix(y_test_secret, test_secret_df['predicted_label'])  
conf_matrix_df = pd.DataFrame(conf_matrix, index=class_dict.values(),  
    columns=class_dict.values())  
conf_matrix_df
```

```
[20]:
```

	T-shirt/top	Trouser	Pullover	Dress	Coat	Sandal	Shirt	\
T-shirt/top	48	96	3	1	6	14	21	
Trouser	8	100	5	3	6	18	31	
Pullover	1	69	26	3	9	11	56	
Dress	1	58	7	29	11	24	33	
Coat	1	109	7	5	22	4	63	
Sandal	2	32	2	2	12	59	32	
Shirt	0	74	4	5	16	15	65	
Sneaker	12	32	2	9	19	30	13	
Bag	0	64	2	3	24	12	42	
Ankle boot	0	30	5	20	14	55	41	

	Sneaker	Bag	Ankle boot
T-shirt/top	4	18	5
Trouser	2	17	1
Pullover	0	9	2
Dress	9	26	1
Coat	0	9	3
Sandal	2	59	9
Shirt	0	16	3
Sneaker	28	61	4
Bag	1	20	0
Ankle boot	4	15	14

```
[21]: grader.check("q6c")
```

```
[21]: q6c results: All test cases passed!
```

0.5.4 Problem 6d: Analysis of Class Accuracies and Confusion Matrix

We observe that the types of classes being misclassified differ significantly from the original test set.

Question: What unusual patterns do you notice in the class accuracies and confusion matrix? What might these patterns suggest about the characteristics of the test set?

From the class accuracy bar plot it can be seen that:

The model exhibits the highest class accuracy for **Bags, Sandals and Shirts**. Conversely, it exhibits the lowest class accuracy for **Trouser and Pullover**.

From the confusion matrix, examining the False Negatives of the Model's Predictions (the value of each column for the row in the Confusion Matrix DataFrame), it can be seen that for **Trouser and Pullover**, the model misclassifies them as **Shirts**. This misclassification can also be seen in **Ankle Boots** being classified as **Sandal or Shirt**, **Bag as Shirt** and **Sneaker as Bag**.

This suggests that the test set might have very different looking images for the same label of item. The model's parameters might have been fitted to characteristics of **Shirts, Bags or Sandals** that the test set's images might closely resemble. This would result in the model misclassifying many items to be **Shirts, Bags or Sandals**.

0.6 Understanding the Issue

To better understand the model's performance, let's examine some misclassifications. These are instances where the model's predictions differ from the true labels.

In the next cell, we will: 1. Identify misclassified examples from the test dataset. 2. Display these examples along with their predicted and true labels.

Pay close attention to whether the misclassifications are concentrated in specific classes or if they seem random.

```
[22]: # Predict labels for the secret test set
predictions_secret = model.predict(X_test_secret_sc)

# Identify misclassified examples
incorrect_secret = predictions_secret != y_test_secret

# Generate labels for misclassified examples with true and predicted labels
labels = [f"True: {true_label}<br>Pred: {pred_label}"
          for true_label, pred_label in zip(y_test_secret[incorrect_secret],
          predictions_secret[incorrect_secret])]

# Display misclassified images with their true and predicted labels
show_images(X_test_secret[incorrect_secret], max_images=5, ncols=5,
            labels=labels, reshape=True)
```

True: T-shirt/top
Pred: Shirt

True: Sandal
Pred: Bag

True: Pullover
Pred: Bag

True: Sandal
Pred: Shirt

True: Shirt
Pred: Sandal



Ah! It seems that all the test images are rotated. This misalignment between the training and test distributions is likely causing the model's poor performance on the rotated test set.

To address this issue, we can explore two potential solutions: 1. **Train a rotation-invariant classifier**: Modify the training data to include rotated images, which allows the model to learn features that are robust to rotation. 2. **Undo the rotation**: Preprocess the test images to align them with the training distribution to enable the current classifier to perform better. 3. **Use test time augmentation**: At the inference time, add augmentation to the data so the model considers augmented data at the test time.

Let's investigate these approaches to improve the model's performance on the rotated test set.

0.7 Problem 7: Solution 1 – Train a Rotation-Invariant Classifier

First, let's try altering our training data to better align with the test distribution. This involves generating augmented data by applying random rotations to the training images. By doing so, we aim to train a model that is more robust to variations in orientation, which is a key characteristic of the rotated test set.

```
[23]: import matplotlib.pyplot as plt

def rotate_image_by_angle(image, angle):
    """
    Rotate an image by a specific angle.
    """
    image = image.reshape(28, 28) # Reshape the image to 28x28
    rotated = rotate(image, angle, reshape=False, cval=0, order=1) # Rotate
    ↪ the image
    return rotated.flatten() # Flatten the rotated image back to 1D

# Visualize a rotated image
plt.imshow(rotate_image_by_angle(X_train[0], 45).reshape(28, 28), cmap='gray_r')
plt.axis('off')
plt.show()
```



0.7.1 Problem 7a: Generating Rotated Training Data

Task: Use the provided rotation function to create augmented training data by randomly rotating the original images.

Hints: - Use NumPy's `random.uniform()` to generate random rotation angles. - Ensure the function returns two NumPy arrays, as `MLPClassifier` expects NumPy arrays as input.

```
[25]: def randomly_rotate_images(images, num_rotations_per_image=5):  
    """  
    Create training data by rotating original images and storing the rotation_  
    ↪ angles as labels.  
    Params:  
    - images: numpy array of shape (n_images, 784)  
    - num_rotations_per_image: int, number of rotations to perform per image  
    Returns:  
    - X_augmented: numpy array of shape (n_images * num_rotations_per_image, ↪  
    ↪ 784)  
    - y_rotations: numpy array of shape (n_images * num_rotations_per_image,)↪  
    """  
    # TODO: Implement this function  
    n_images = images.shape[0]  
    X_augmented = np.zeros((n_images * num_rotations_per_image, 784))  
    y_rotations = np.zeros((n_images * num_rotations_per_image,))
```

```

for i in range(n_images):
    for j in range(num_rotations_per_image):
        angle = np.random.uniform(low = 0, high = 360)
        X_augmented[i * num_rotations_per_image + j] =
↪rotate_image_by_angle(images[i], angle)
        y_rotations[i * num_rotations_per_image + j] = angle
    return X_augmented, y_rotations

X_train = X_train[:3000]
y_train = y_train[:3000]

num_rotations_per_image = 4
X_train_rotated, y_train_rotations = randomly_rotate_images(X_train,
↪num_rotations_per_image)
y_train_augmented = np.array([y for label in y_train for y in
↪[label]*num_rotations_per_image])
print(X_train_rotated.shape)
print(y_train_augmented.shape)

show_images(X_train_rotated, max_images=5, ncols=5, labels=y_train_augmented,
↪reshape=True)

```

(12000, 784)

(12000,)

Sneaker

Sneaker

Sneaker

Sneaker

Dress



```
[26]: grader.check("q7a")
```

[26]: q7a results: All test cases passed!

0.7.2 Problem 7b: Training a Rotation-Invariant Classifier

Now that we have augmented our training data with rotated images, let's train a new `MLPClassifier` on this dataset.

Task: Train a new `MLPClassifier` using the rotated training images `X_train_rotated_sc` and the corresponding labels `y_train_augmented`.

Steps: 1. Scale the rotated training data `X_train_rotated` using the `scaler` to create `X_train_rotated_sc`. 2. Train the `MLPClassifier` on the scaled data `X_train_rotated_sc` and labels `y_train_augmented`.

Important for grading: - Save your trained model as `model_rotated`. - Use `random_state=42` when instantiating the `MLPClassifier` for reproducibility. - Ensure your loss plot is displayed by making the plotting code the last expression in the cell or explicitly calling `plt.show()`.

Hint: Refer to [Scikit-learn's MLPClassifier documentation](#) for additional guidance.

```
[27]: X_train_rotated_sc = scaler.transform(X_train_rotated)
      X_test_secret_sc = scaler.transform(X_test_secret)
```

```

if (load_saved_models or IS_GRADING_ENV) and os.path.
    exists('mlp_fashionmnist_rotated_model.joblib'):
    model_rotated = joblib.load("mlp_fashionmnist_rotated_model.joblib")
else:
    # TODO: initialize and train a new model on the rotated images
    model_rotated = MLPClassifier(hidden_layer_sizes=(100, 50),max_iter=100,
    tol=1e-3,random_state = SEED)
    model_rotated.fit(X_train_rotated_sc,y_train_augmented)

    if save_models:
        joblib.dump(model_rotated, "mlp_fashionmnist_rotated_model.joblib")
        print("Rotated model saved to disk.")

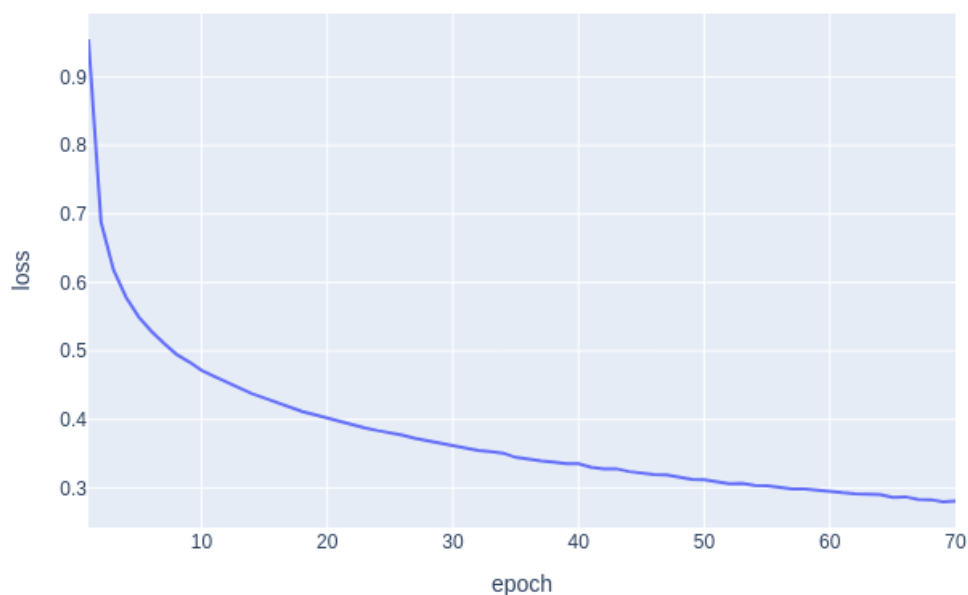
print(f"Training accuracy (training on rotated images): {model_rotated.
    score(X_train_rotated_sc, y_train_augmented):.3f}")
print(f"Test accuracy (original): {model.score(X_test_sc, y_test):.3f}")
print(f"Test accuracy (training on rotated images): {model_rotated.
    score(X_test_secret_sc, y_test_secret):.3f}")

loss_df = pd.DataFrame({
    'epoch': range(1, len(model_rotated.loss_curve_) + 1),
    'loss': model_rotated.loss_curve_
})
loss_df.plot(x='epoch', y='loss', title="Training Error")

```

Training accuracy (training on rotated images): 0.904
 Test accuracy (original): 0.884
 Test accuracy (training on rotated images): 0.773

Training Error



```
[28]: grader.check("q7b")
```

[28]: q7b results: All test cases passed!

0.8 Problem 8: Solution 2 – Undo the Rotations in the Secret Test Set

In this approach, we aim to align the test distribution with the training distribution by building a regression model to predict the rotation angle of an image. Once the rotation angle is predicted, we can rotate the image back to its original orientation (angle 0) before feeding it into the classifier.

0.8.1 Problem 8a: Predicting Rotation Angles

Task: Train a multi-layer perceptron (MLPRegressor) to predict the rotation angle of an image. Make sure you follow the same data processing steps from previous sections. We have provided the model and hyperparameters for you.

Instructions:

- Save your trained MLPRegressor as `model_rotation_regressor`.
- Use `random_state=42` when instantiating your MLPRegressor to ensure reproducibility.
- If your loss plot does not display, ensure the code for plotting the loss is the *last expression* in your Jupyter cell, or explicitly call `plt.show()`.

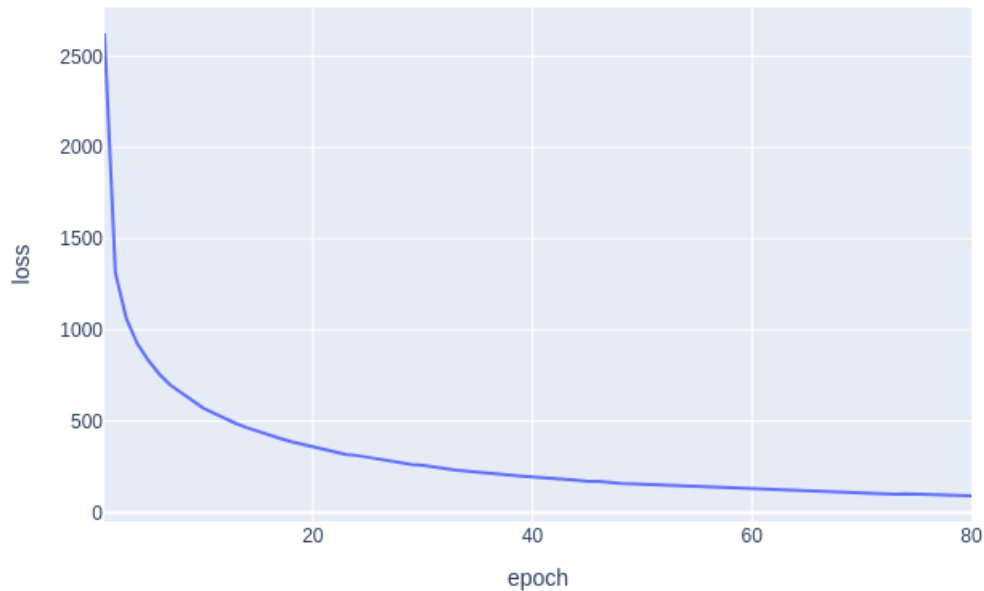
Hints:

1. Refer to [Scikit-learn's MLPRegressor documentation](#) for guidance on configuring the model.
2. Train the model for enough iterations to observe convergence. The `early_stopping`

parameter can be helpful to stop training automatically once convergence is reached. If the model trains for too many iterations without converging, consider adjusting other MLPRegressor parameters.

```
[29]: if (load_saved_models or IS_GRADING_ENV) and os.path.  
      ↪exists('model_rotation_regression.joblib'):  
      model_rotation_regressor = joblib.load("model_rotation_regression.joblib")  
      model_rotation_regression = model_rotation_regressor  
else:  
    # Train a larger regression model (MLP) to predict rotation angles  
    model_rotation_regression = MLPRegressor(  
        hidden_layer_sizes=(256, 128),  
        activation='relu',  
        solver='adam',  
        max_iter=100,  
        random_state=SEED,  
        early_stopping=True,  
        verbose=True  
    )  
    # TODO: train a MLP regressor to predict rotation angles  
    model_rotation_regression = model_rotation_regression.fit(X_train_rotated_sc, ↪  
    ↪y_train_rotations)  
    # Save model_rotation_regression and scaler  
    if save_models:  
        joblib.dump(model_rotation_regression, "model_rotation_regression.joblib")  
        joblib.dump(scaler, "scaler.joblib")  
        print("Rotation regression model and scaler saved to disk.")  
  
loss_df = pd.DataFrame({  
    'epoch': range(1, len(model_rotation_regression.loss_curve_) + 1),  
    'loss': model_rotation_regression.loss_curve_  
})  
loss_df.plot(x='epoch', y='loss', title="Training Loss")
```

Training Loss



```
[30]: grader.check("q8a")
```

[30]: q8a results: All test cases passed!

0.8.2 Problem 8b: Evaluating the Rotation Angle Prediction Model

In this task, we will evaluate the performance of our `model_rotation_regression` in predicting the rotation angles of images.

Steps: 1. Use the trained `model_rotation_regression` to predict the rotation angles for the images in `X_test_rotated_sc`. 2. Calculate the **Mean Squared Error (MSE)** and **Root Mean Squared Error (RMSE)** of the predictions.

Hints: - Use the `mean_squared_error` function from `sklearn.metrics` to compute the MSE. - To calculate the RMSE, take the square root of the MSE. You can use `np.sqrt()` for this purpose.

```
[31]: # TODO: Use the model_rotation_regression to predict the rotation angles on
      ↪ X_test_rotated_sc
X_test_rotated, y_test_rotations = randomly_rotate_images(X_test,
      ↪ num_rotations_per_image)
X_test_rotated_sc = scaler.transform(X_test_rotated)

y_pred_angles = model_rotation_regression.predict(X_test_rotated_sc)
```

```

mse = model_rotation_regression.score(X_test_rotated_sc, y_test_rotations)
rmse = np.sqrt(mse)
print(f"Test MSE: {mse:.2f}")
print(f"Test RMSE: {rmse:.2f} degrees")

# Show some predictions
print("\nSample predictions:")
for i in range(5):
    print(f"True rotation: {y_test_rotations[i]:.1f}°, Predicted: {y_pred_angles[i]:.1f}°")

print("-----")
print("-----")
print(f"Test MSE (angle): {mse:.2f}")
print(f"Test RMSE: {rmse:.2f}°")
print("-----")
print("-----")

```

Test MSE: 0.89

Test RMSE: 0.94 degrees

Sample predictions:

True rotation: 235.6°, Predicted: 228.9°

True rotation: 28.8°, Predicted: 22.6°

True rotation: 87.2°, Predicted: 69.5°

True rotation: 278.5°, Predicted: 263.5°

True rotation: 190.3°, Predicted: 162.6°

Test MSE (angle): 0.89

Test RMSE: 0.94°

```
[32]: grader.check("q8b")
```

[32]: q8b results: All test cases passed!

0.8.3 Relationship between the Predicted vs True Rotation Angle

Let's visualize how well `model_rotation_regression` was able to predict the true rotation angle.

```

[33]: # Create scatter plot of predictions vs actual
scatter_fig = go.Figure()
scatter_fig.add_trace(
    go.Scatter(
        x=y_test_rotations[:200], # First 200 samples for clarity

```

```

        y=y_pred_angles[:200],
        mode='markers',
        name='Predictions',
        marker=dict(size=6, opacity=0.7)
    )
)

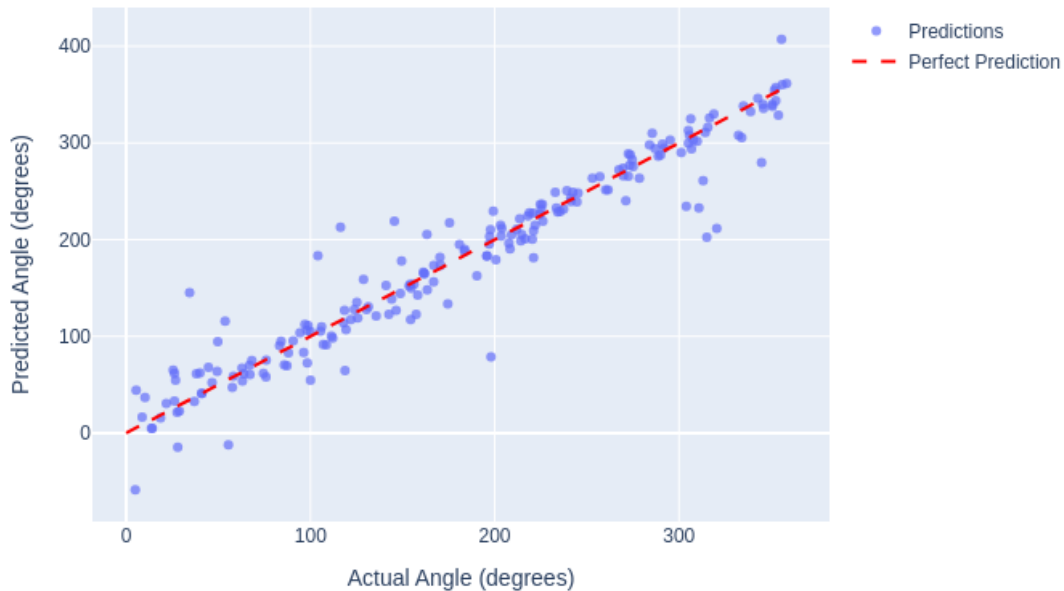
# Add perfect prediction line
max_angle = max(y_test_rotations[:200])
scatter_fig.add_trace(
    go.Scatter(
        x=[0, max_angle],
        y=[0, max_angle],
        mode='lines',
        name='Perfect Prediction',
        line=dict(dash='dash', color='red')
    )
)

scatter_fig.update_layout(
    title='Predicted vs Actual Rotation Angles',
    xaxis_title='Actual Angle (degrees)',
    yaxis_title='Predicted Angle (degrees)',
    width=600,
    height=500
)

scatter_fig.show()

```

Predicted vs Actual Rotation Angles



0.8.4 Test model_rotation_regression in Action

Next, let's actually try to use the rotation angles that `model_rotation_regression` predicts to unrotate the images back to their original state.

```
[34]: # Demonstrate unrotation using predicted angles
def unrotate_image(rotated_image, predicted_angle):
    """
    Unrotate an image by rotating it by the negative of the predicted angle.
    """
    return rotate_image_by_angle(rotated_image, -predicted_angle)

# Test unrotation on a sample
sample_idx = 0
original_test_image = X_test[sample_idx // 1] # Get original unrotated image
rotated_test_image = X_test_rotated[sample_idx] # Get the rotated image
true_angle = y_test_rotations[sample_idx] # Get the true angle the image was
↳ rotated by
predicted_angle = y_pred_angles[sample_idx] # Get the model's prediction
unrotated_image = unrotate_image(rotated_test_image, predicted_angle)

# Prepare images and titles for display
```

```

images = [
    original_test_image.reshape(28, 28)[::-1],
    rotated_test_image.reshape(28, 28)[::-1],
    unrotated_image.reshape(28, 28)[::-1]
]
titles = [
    'Original',
    f'Rotated by {true_angle:.1f}°',
    f'After unrotating the image with the model\'s prediction (pred:␣
↪{predicted_angle:.1f}°)'
]

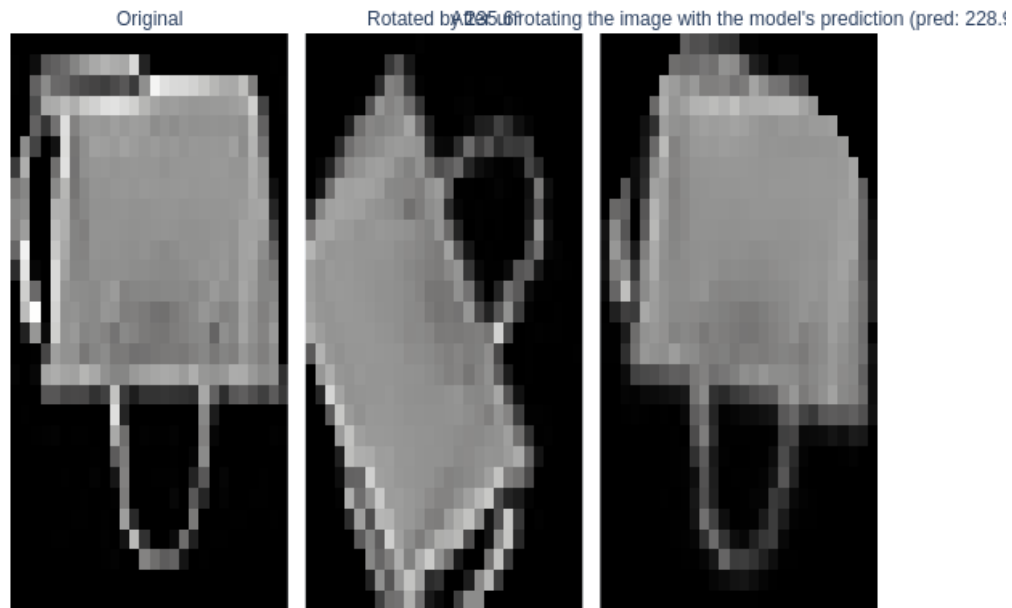
# Create a subplot grid using plotly express imshow
fig = px.imshow(
    np.stack(images),
    facet_col=0,
    facet_col_wrap=3,
    color_continuous_scale='gray',
    aspect='auto'
)

# Update facet titles
for i, title in enumerate(titles):
    fig.layout.annotations[i]['text'] = title

fig.update_layout(
    height=300,
    width=1200,
    title_text="Image Rotation Prediction and Correction",
    coloraxis_showscale=False
)
fig.update_xaxes(showticklabels=False, showgrid=False, zeroline=False)
fig.update_yaxes(showticklabels=False, showgrid=False, zeroline=False)
fig.show()

```

Image Rotation Prediction and Correction



0.8.5 Problem 8c: Unrotating Images and Evaluating the Classifier

Task: 1. Unrotate the images in `X_test_secret` by predicting their rotation angles.

- Scale the secret test set ``X_test_secret`` using the ``scaler`` and save the scaled data into ``X_test_scaled``.
- Use the trained ``model_rotation_regression`` to predict the rotation angles of the images (``y``).
- Use the predicted rotation angles to *unrotate* the images in ``X_test_secret``. Save the unrotated images into ``X_test_unrotated``.
- Scale the unrotated images using the ``scaler`` and save the scaled data into ``X_test_unrotated_sc``.

2. Evaluate the classifier on the unrotated images.

- Use the original `MLPClassifier` (`model`) to make predictions on `X_test_unrotated_sc``.
- Check which images are correctly classified.

Expected Output: - Add two new columns to `test_secret_df``: - `"unrotated_predictions"`: Contains the predicted classes for the unrotated images. - `"unrotated_correct"`: Boolean values indicating whether the predicted class matches the ground truth.

Hints: - Use `scaler.transform` to scale the data before using `model_rotation_regression` to predict rotation angles and after unrotating the images. - Ensure that the unrotated images are properly scaled before passing them to `model` for classification.

```
[35]: # TODO: Scale the secret test set and use model_rotation_regression to unrotate
      ↪the images in the secret test set
X_test_secret_scaled = scaler.transform(X_test_secret)
y_pred_angles = model_rotation_regression.predict(X_test_secret_scaled)
X_test_unrotated = np.array([unrotate_image(image, angle) for image, angle in
      ↪zip(X_test_secret, y_pred_angles)])
X_test_unrotated_sc = scaler.transform(X_test_unrotated)
```

```
[36]: grader.check("q8ci")
```

[36]: q8ci results: All test cases passed!

```
[37]: # TODO: Make new predictions using the original MLPClassifier model and check
      ↪which images are correctly classified
test_secret_df["unrotated_prediction"] = model.predict(X_test_unrotated_sc)
test_secret_df["unrotated_correct"] = test_secret_df["unrotated_prediction"] ==
      ↪test_secret_df["label"]

print(f"Test accuracy: {test_secret_df.correct.mean():.3f}")
print(f"Unrotated Test accuracy: {test_secret_df.unrotated_correct.mean():.3f}")
```

Test accuracy: 0.205

Unrotated Test accuracy: 0.559

```
[38]: grader.check("q8cii")
```

[38]: q8cii results: All test cases passed!

0.8.6 Problem 8d: Comparing Data Augmentation and Rotation Correction

Question: Why might the rotation correction approach (Solution 2) perform worse than the data augmentation approach (Solution 1)?

The rotation correction approach is affected by how well trained the rotation prediction model is. If the rotation is over or under corrected, the final prediction done by the model will still be inaccurate due to the residual rotation.

On the other hand, training a rotation invariant model reduces the impact of the rotation of the images on the accuracy.

Let's compare our three strategies for handling rotated images: doing nothing (baseline), training a model on rotated data, and a two-step pipeline that predicts orientation then un-rotates before classification.

1. First, we'll make predictions using our 3 strategies.
2. Then, we'll store the accuracies of all 3 methods into a dataframe
3. Lastly, we'll plot the accuracies in a color-coded bar chart and print the accuracies.

Take a look at which method best overcomes the rotation effects.

```

[39]: # Compare all three approaches
baseline_accuracy = model.score(X_test_secret_sc, y_test_secret) # method:
↳baseline
rotated_training_accuracy = model_rotated.score(X_test_secret_sc,
↳y_test_secret) # method: train on rotated images
unrotated_accuracy = model.score(X_test_unrotated_sc, y_test_secret) # method:
↳predict & unrotate, then classify the images we tried to unrotate

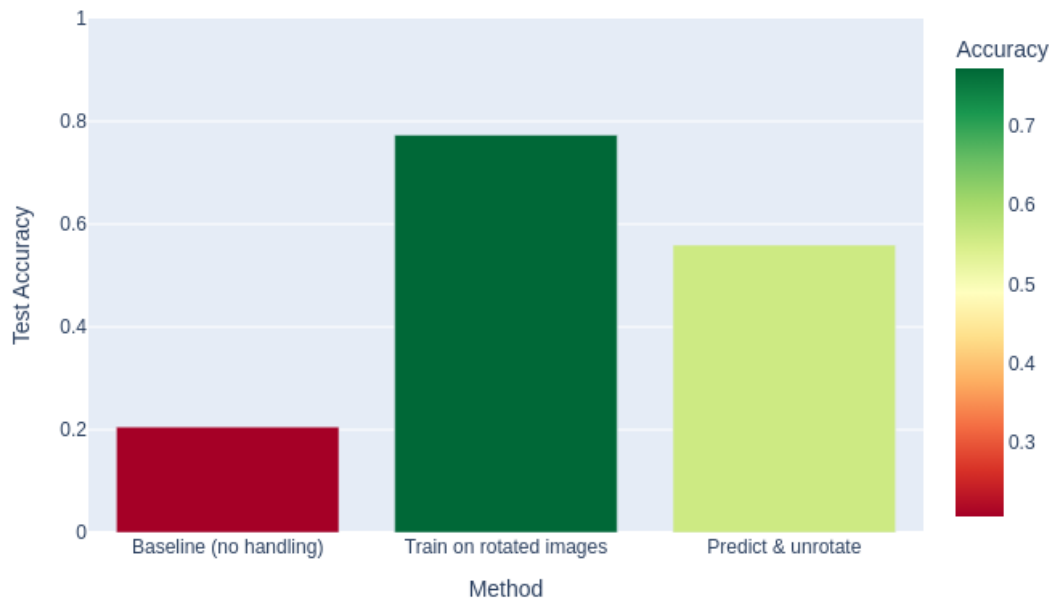
# Create comparison DataFrame
comparison_df = pd.DataFrame({
    'Method': ['Baseline (no handling)', 'Train on rotated images', 'Predict &
↳unrotate'],
    'Accuracy': [baseline_accuracy, rotated_training_accuracy,
↳unrotated_accuracy]
})

# Plot comparison
fig = px.bar(
    comparison_df,
    x='Method',
    y='Accuracy',
    title='Comparison of Rotation Handling Methods',
    color='Accuracy',
    color_continuous_scale='RdYlGn'
)
fig.update_layout(
    xaxis_title="Method",
    yaxis_title="Test Accuracy",
    yaxis=dict(range=[0, 1])
)
fig.show()

print("\n=== Summary of All Methods ===")
for _, row in comparison_df.iterrows():
    print(f"{row['Method']}: {row['Accuracy']:.3f}")

```

Comparison of Rotation Handling Methods



=== Summary of All Methods ===

Baseline (no handling): 0.205

Train on rotated images: 0.773

Predict & unrotate: 0.559

0.9 Problem 9: Solution 3 – Enhancing Accuracy with Test Time Augmentations

Test Time Augmentation (TTA) is a powerful technique to improve model performance during inference by applying data transformations to the test data. For example, transformations such as rotations, flips, or other augmentations can be applied to the test images before passing them to the model. The predictions from these augmented images can then be aggregated to make a more robust final prediction.

Task: 1. Use test time augmentations to improve the accuracy of the original classifier. Your final accuracy must reach at least **58%**. Modify the `test_time_augmentation` function (and add any additional helper functions or transformations) as needed.

a. You are allowed to use the rotation prediction model (``model_rotation_regression``) and can n

b. You may reuse functions defined earlier in the notebook to implement your test-time augmenta

Important for grading: - You must use the original `MLPClassifier` (`model`) for classification

and cannot train any additional models.

```
[40]: # TODO: Write a function (or functions!) that can be used to improve the
      ↪ accuracy of the original MLPClassifier model using test time augmentations
def flat_to_2d(x_flat: np.ndarray) -> np.ndarray:
    x = np.asarray(x_flat, dtype=np.float32)
    return x.reshape(28, 28)

def flip(image, flip_h = True, flip_v = False):
    tmp = []
    outs = []
    if flip_h: tmp.append(np.fliplr(image))
    if flip_v: tmp.append(np.flipud(image))
    for _ in tmp:
        outs.append(_.flatten())
    return outs

# Accuracy Bump due to: Scale once at end, augment on raw data
# Feel free to add additional functions, arguments, etc. as needed!
def test_time_augmentation(model, scaler, image):
    """
    Predict a label using test-time augmentation
    Params:
        - model: the MLPClassifier model
        - scaler: the StandardScaler used to scale the images
        - image: the image to be augmented
    Returns:
        - prediction: the predicted label
    """
    variants = []
    rotations = (-20, -10, -5, -2, 2, 5, 10, 20)
    variants.append(image)
    angle_input = scaler.transform(image[np.newaxis, :])
    y_pred_angle = model_rotation_regression.predict(angle_input)[0]
    image = unrotate_image(image, y_pred_angle)
    variants.append(image)
    for i in range(len(rotations)):
        angle = rotations[i]
        variants.append(rotate_image_by_angle(image, angle))

    flip_outs = flip(flat_to_2d(image), flip_h = True)
    for out in flip_outs:
        variants.append(out)

    X_tta = np.stack(variants, axis=0)
    X_tta_scaled = scaler.transform(X_tta)
    predictions = model.predict(X_tta_scaled)
```

```

from collections import Counter
most_common = Counter(predictions).most_common(1)
final_prediction = most_common[0][0]

return final_prediction

# Make a copy of the test secret dataframe and apply the test time augmentation
↳to the images to get new predictions!
part_2_df = test_secret_df.copy()
part_2_df["image"] = part_2_df["image"].apply(lambda x: np.array(x).reshape(-1))
part_2_df["prediction"] = part_2_df["image"].apply(lambda x:
↳test_time_augmentation(model, scaler, x))

# Check the accuracy of the new predictions
correct = part_2_df["prediction"] == part_2_df["label"]
print("Accuracy:", correct.mean())

```

Accuracy: 0.5835

1 Before you submit, ensure save_models is true!

Important for grading: Make sure you set these variables to True at the top of your notebook before submitting.

```

[41]: assert save_models and load_saved_models, "save_models and load_saved_models
↳must be True"

assert os.path.exists('mlp_fashionmnist_model.joblib'), "mlp_fashionmnist_model.
↳joblib should exist"
assert os.path.exists('price_model.joblib'), "price_model.joblib should exist"
assert os.path.exists('mlp_fashionmnist_rotated_model.joblib'),
↳"mlp_fashionmnist_rotated_model.joblib should exist"
assert os.path.exists('model_rotation_regression.joblib'),
↳"model_rotation_regression.joblib should exist"

```

1.1 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a zip file for you to submit. **Please save before exporting!**

```

[42]: ## Use this cell if you are running the notebook in Google Colab to install the
↳necessary dependencies, this may take a few minutes
if IS_COLAB:
    !apt-get install -y texlive texlive-xetex pandoc

```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc-data poppler-data preview-latex-style
rake ruby ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
rubygems-integration tlutils teckit tex-common tex-gyre texlive-base
texlive-binaries texlive-fonts-recommended texlive-latex-base
texlive-latex-extra texlive-latex-recommended texlive-pictures
texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
texlive-luatex pandoc-citeproc context wkhtmltopdf librsvg2-bin groff ghc
nodejs php python libjs-mathjax libjs-katex citation-style-language-styles
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-pstricks
dot2tex prerex texlive-pictures-doc vprerex default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc pandoc-data poppler-data
preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-webrick
ruby-xmlrpc ruby3.0 rubygems-integration tlutils teckit tex-common tex-gyre
texlive texlive-base texlive-binaries texlive-fonts-recommended
texlive-latex-base texlive-latex-extra texlive-latex-recommended
texlive-pictures texlive-plain-generic texlive-xetex tipa xfonts-encodings
xfonts-utils

0 upgraded, 58 newly installed, 0 to remove and 35 not upgraded.

Need to get 202 MB of archives.

After this operation, 728 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]
Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]
Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]
Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]
Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.12 [753 kB]
Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]
Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]
Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]
Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.12 [5,031 kB]
Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]
Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [115 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm-extensions0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [25.1 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.2 [228 kB]

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 [5,100 B]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
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 2.5.11+ds1-1 [421 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
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 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
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 1:1.0.5-0ubuntu2 [578 kB]
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 1:7.7+6build2 [94.6 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 pandoc-data all
 2.9.2.1-3ubuntu2 [81.8 kB]
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 2.9.2.1-3ubuntu2 [20.3 MB]
 Get:42 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
 all 12.2-1ubuntu1 [185 kB]
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 1.41-4build2 [61.3 kB]
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 2.5.11+ds1-1 [699 kB]
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 20180621-3.1 [6,209 kB]
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 binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
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 2021.20220204-1 [21.0 MB]
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 recommended all 2021.20220204-1 [4,972 kB]
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 all 2021.20220204-1 [1,128 kB]

```

Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
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2021.20220204-1 [14.3 kB]
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1:1.8.16-2 [207 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
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all 2021.20220204-1 [8,720 kB]
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all 2021.20220204-1 [13.9 MB]
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generic all 2021.20220204-1 [27.5 MB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:58 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 202 MB in 4s (48.7 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126435 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.12_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.

```

```

Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.12_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcmark-gfm0.29.0.gfm.3:amd64.
Preparing to unpack .../17-libcmark-gfm0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcmark-gfm-
extensions0.29.0.gfm.3:amd64.
Preparing to unpack .../18-libcmark-gfm-
extensions0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../19-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../20-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../21-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../22-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...

```

```

Selecting previously unselected package ruby3.0.
Preparing to unpack .../23-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../24-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../25-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../26-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../27-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../28-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../29-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../30-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../31-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../32-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../33-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../34-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../35-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../36-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../37-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...

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Selecting previously unselected package lmodern.
 Preparing to unpack .../38-lmodern_2.004.5-6.1_all.deb ...
 Unpacking lmodern (2.004.5-6.1) ...
 Selecting previously unselected package pandoc-data.
 Preparing to unpack .../39-pandoc-data_2.9.2.1-3ubuntu2_all.deb ...
 Unpacking pandoc-data (2.9.2.1-3ubuntu2) ...
 Selecting previously unselected package pandoc.
 Preparing to unpack .../40-pandoc_2.9.2.1-3ubuntu2_amd64.deb ...
 Unpacking pandoc (2.9.2.1-3ubuntu2) ...
 Selecting previously unselected package preview-latex-style.
 Preparing to unpack .../41-preview-latex-style_12.2-1ubuntu1_all.deb ...
 Unpacking preview-latex-style (12.2-1ubuntu1) ...
 Selecting previously unselected package t1utils.
 Preparing to unpack .../42-t1utils_1.41-4build2_amd64.deb ...
 Unpacking t1utils (1.41-4build2) ...
 Selecting previously unselected package teckit.
 Preparing to unpack .../43-teckit_2.5.11+ds1-1_amd64.deb ...
 Unpacking teckit (2.5.11+ds1-1) ...
 Selecting previously unselected package tex-gyre.
 Preparing to unpack .../44-tex-gyre_20180621-3.1_all.deb ...
 Unpacking tex-gyre (20180621-3.1) ...
 Selecting previously unselected package texlive-binaries.
 Preparing to unpack .../45-texlive-binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
 Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
 Selecting previously unselected package texlive-base.
 Preparing to unpack .../46-texlive-base_2021.20220204-1_all.deb ...
 Unpacking texlive-base (2021.20220204-1) ...
 Selecting previously unselected package texlive-fonts-recommended.
 Preparing to unpack .../47-texlive-fonts-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-fonts-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-base.
 Preparing to unpack .../48-texlive-latex-base_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-base (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-recommended.
 Preparing to unpack .../49-texlive-latex-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive.
 Preparing to unpack .../50-texlive_2021.20220204-1_all.deb ...
 Unpacking texlive (2021.20220204-1) ...
 Selecting previously unselected package libfontbox-java.
 Preparing to unpack .../51-libfontbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libfontbox-java (1:1.8.16-2) ...
 Selecting previously unselected package libpdfbox-java.
 Preparing to unpack .../52-libpdfbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libpdfbox-java (1:1.8.16-2) ...
 Selecting previously unselected package texlive-pictures.
 Preparing to unpack .../53-texlive-pictures_2021.20220204-1_all.deb ...

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Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../54-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../55-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../56-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../57-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up pandoc-data (2.9.2.1-3ubuntu2) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...

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```

Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up pandoc (2.9.2.1-3ubuntu2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

```

```

Processing triggers for tex-common (6.17) ...
Running upmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.

```

```

[43]: # Save your notebook first, then run this cell to export your submission.
grader.export(run_tests=True, files=['mlp_fashionmnist_model.joblib',
↳ 'price_model.joblib', 'mlp_fashionmnist_rotated_model.joblib',
↳ 'model_rotation_regression.joblib'])

```

Running your submission against local test cases...

Your submission received the following results when run against available test cases:

```
q5a results: All test cases passed!
q5b results: All test cases passed!
q5d results: All test cases passed!
q6a results: All test cases passed!
q6c results: All test cases passed!
q7a results: All test cases passed!
q7b results: All test cases passed!
q8a results: All test cases passed!
q8b results: All test cases passed!
q8ci results: All test cases passed!
q8cii results: All test cases passed!
<IPython.core.display.HTML object>
```